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Pentagonal Number Theorem

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1 Partitions

Definition 1. A *partition* of a positive integer n, also called an integer partition, is a way of writing n as a sum of positive integers. Two sums that differ only in the order of their summands are considered the same partition.

The *partition function* $p(n)$ represents the number of possible partitions of a natural number n.

Example 2. $5 = 4+1 = 3+2 = 3+1+1 = 2+2+1 = 2+1+1+1 = 1+1+1+1+1$
$p(5) = 7$

Lemma 3. $p_{c}ount, pGenFun, coeff_{p}GenFun_{e}q_{pc}ount, pGenFun_{e}q_{p}roddef : partitionThegeneratingfunctio$
$\prod_{k=1}^{\infty}(1 + x^k + x^{2k} + \ldots) = \prod_{k=1}^{\infty}(1 - x^k)^{-1}$.

Proof. We work in the ring of formal power series $\mathbb{Z}[[x]]$.

Step 1. For each fixed $k \geq 1$, the geometric series identity

$$1 + x^k + x^{2k} + \cdots = \sum_{j=0}^{\infty} x^{jk} = (1 - x^k)^{-1}$$

holds as formal power series. This gives the second equality.

Step 2. Expand the product $\prod_{k=1}^{\infty}(1 + x^k + x^{2k} + \cdots)$. Each term in the expansion is obtained by selecting, for every $k \geq 1$, an exponent $j_k \in \mathbb{Z}_{\geq 0}$ (with $j_k = 0$ for all but finitely many k) and multiplying the chosen monomials. Such a selection contributes a single term

$$\prod_k x^{j_k \cdot k} = x^{\sum_k j_k \cdot k}.$$

Step 3. The coefficient of x^n is therefore the number of tuples $(j_1, j_2, \ldots)$ with $\sum_k k \cdot j_k = n$.

Step 4. Such a tuple corresponds bijectively to a partition of n: j_k is the multiplicity of the part k. Hence the coefficient of x^n equals $p(n)$, and the first equality follows. $\square$

Definition 4 (Partitions of n into distinct parts)**.** $\mathrm{all}_d istinct_p artitions, even_d istinct_p artitions, odd_d istinct_p art$

Let $n \in \mathbb{N}$. We define the following sets of partitions of n into distinct positive parts:

$$\mathcal{P}(n) = \left\{ S \subseteq \{1, \ldots, n\} \;\middle|\; \sum_{s \in S} s = n \right\},$$
$$\mathcal{P}_{\text{even}}(n) = \{S \in \mathcal{P}(n) \mid |S| \equiv 0 \pmod 2\},$$
$$\mathcal{P}_{\text{odd}}(n) = \{S \in \mathcal{P}(n) \mid |S| \equiv 1 \pmod 2\}.$$

We further define the number of even and odd partitions of n as $p_e(n) = |\mathcal{P}_{\text{even}}(n)|$ and $p_o(n) = |\mathcal{P}_{\text{odd}}(n)|$.

Lemma 5. *$coeff_p rod_e q_p e_s ub_p o, coef f_p rod_e q_s igned_p artition_s um, signed_p artition_s um_e q_p e_s ub_p odef : distinct -$ $x^k) = \sum_{s=0}^{\infty} \sum_{k_1<k_2<\cdots<k_s} (-1)^s \, x^{k_1+k_2+\ldots+k_s} = \sum_{n=0}^{\infty} c_n \, x^n where c_0 = 1$ and $c_n = p_e(n) - p_o(n)$ for $n \geq 1$.*

Proof. We expand the infinite product as a formal power series.

Step 1: Expansion as a sum over finite subsets. For each $k \geq 1$, the factor $(1 - x^k)$ contributes either 1 or $-x^k$ when expanding the product. A choice across all factors corresponds to a finite subset

$$K = \{k_1 < k_2 < \cdots < k_s\} \subseteq \mathbb{Z}_{\geq 1}$$

(the indices for which $-x^k$ was selected). The contribution is

$$\prod_{k \in K} (-x^k) = (-1)^{|K|} \, x^{\sum_{k \in K} k} = (-1)^s \, x^{k_1+\cdots+k_s}.$$

Summing over all such K yields the first equality.

Step 2: Grouping by total exponent. A subset K of size s summing to n is exactly a partition $S \in \mathcal{P}(n)$ with $|S| = s$. Hence the coefficient of x^n equals

$$c_n = \sum_{S \in \mathcal{P}(n)} (-1)^{|S|}.$$

Step 3: $c_0 = 1$. Only $S = \emptyset$ partitions 0 into distinct positive parts, contributing $(-1)^0 = 1$.

Step 4: $c_n = p_e(n) - p_o(n)$ for $n \geq 1$. Splitting the sum by parity of $|S|$:

$$c_n = \sum_{\substack{S \in \mathcal{P}(n) \\ |S| \text{ even}}} 1 \; - \sum_{\substack{S \in \mathcal{P}(n) \\ |S| \text{ odd}}} 1 = |\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)| = p_e(n) - p_o(n).$$

$\square$

Example 6. *We compute $\prod_{i=1}^{20}(1 - x^i) + o(x^{20})$.*

The answer is $1 - x - x^2 + x^5 + x^7 - x^{12} - x^{15} + O\left(x^{20}\right)$.

n		$p_o(n)$	$p_e(n)$
1	1	1	0
2	2	1	0
3	$3 = 2 + 1$	1	1
4	$4 = 3 + 1$	1	1
5	$5 = 4 + 1 = 3 + 2$	1	2
6	$6 = 5 + 1 = 4 + 2 = 3 + 2 + 1$	2	2
7	$7 = 6 + 1 = 5 + 2 = 4 + 3 = 4 + 2 + 1$	2	3

2 Pentagonal number theorem

Theorem 7 (Euler)**.** *$coeff_p rod_p entagonal_z ero, coeff_p rod_p entagonal_n onpent, coeff_p rod_p entagonal_m inus, coef$*
$x^i) = 1 + \sum_{k=1}^{\infty}(-1)^k \left(x^{(3k^2-k)/2} + x^{(3k^2+k)/2}\right) = \sum_{k\in\mathbb{Z}}(-1)^k\, x^{(3k^2-k)/2}$.

Proof. lemma:expansion-product, lemma:pe-minus-po-closed-form The proof combines the previous expansion lemma with the closed-form computation of $p_e(n) - p_o(n)$ via Franklin's involution.

Step 1: Coefficient identification. By Lemma **??**,

$$\prod_{i=1}^{\infty}(1 - x^i) = \sum_{n=0}^{\infty} c_n\, x^n,$$

where $c_0 = 1$ and $c_n = p_e(n) - p_o(n)$ for $n \geq 1$.

Step 2: Closed form for c_n. By Lemma **??** below,

$$p_e(n) - p_o(n) = \begin{cases} (-1)^k & \text{if } n = (3k^2 - k)/2 \text{ for some } k \in \mathbb{Z}_{\geq 1}, \\ (-1)^k & \text{if } n = (3k^2 + k)/2 \text{ for some } k \in \mathbb{Z}_{\geq 1}, \\ 0 & \text{otherwise.} \end{cases}$$

Step 3: Reassembling the sum. The values of n where $c_n \neq 0$ are exactly the (generalized) pentagonal numbers $n = (3k^2 \pm k)/2$ for $k \geq 1$, plus $n = 0$. Each contributes $(-1)^k$. Hence

$$\prod_{i=1}^{\infty}(1 - x^i) = 1 + \sum_{k=1}^{\infty}(-1)^k \left(x^{(3k^2-k)/2} + x^{(3k^2+k)/2}\right).$$

Step 4: Bilateral sum reformulation. Reindex the second family by $k \mapsto -k$. For $k \in \mathbb{Z}_{\geq 1}$, set $k' = -k$, so $k' \in \mathbb{Z}_{\leq -1}$ and

$$\frac{3(k')^2 - k'}{2} = \frac{3k^2 + k}{2}, \qquad (-1)^k = (-1)^{|k'|} = (-1)^{k'}.$$

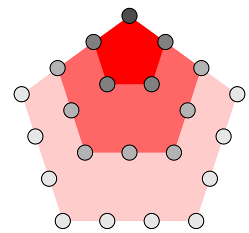
The $k = 0$ term gives $(-1)^0\, x^0 = 1$, matching the leading 1. Combining the three contributions:

$$\sum_{k\in\mathbb{Z}_{\geq 1}}(-1)^k x^{(3k^2-k)/2} + (-1)^0 x^0 + \sum_{k'\in\mathbb{Z}_{\leq -1}}(-1)^{k'} x^{(3(k')^2-k')/2} = \sum_{k\in\mathbb{Z}}(-1)^k x^{(3k^2-k)/2}.$$

□

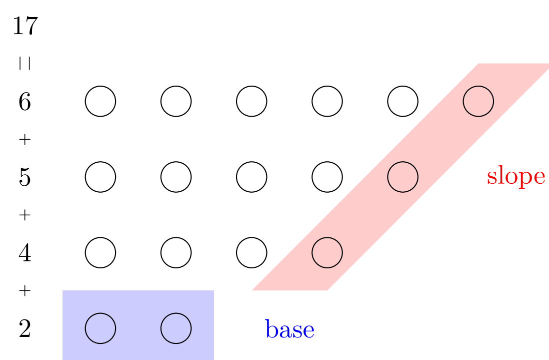
Remark 8 (Pentagonal numbers)**.**

k	-3	-2	-1	0	1	2	3
$\frac{3k^2-k}{2}$	15	7	2	0	1	5	12



2.1 Proof

Let n be a positive integer. Consider a diagram of a partition of n into different parts.



Definition 9 (Base and slope)**.** base, partBase, partMax, partSlope, partSlopeSetdef:distinct-partitionsConsider a nonempty partition $S \subseteq \{1, \ldots, n\}$ of n into different parts.
We define the *base* $b \in S$ as the smallest element of the partition, $b = \min(S)$.
Let $\max(S)$ denote the largest element of S. We define the *slope* s as:

$$s = \max\{k \geq 1 : \max(S), \max(S) - 1, \max(S) - 2, \ldots, \max(S) - k + 1 \in S\}.$$

and the *slope set* D as $\{\max(S), \max(S) - 1, \ldots, \max(S) - s + 1\}$.

Definition 10. DPalpha, DPbeta, DPspecialdef:base-slope, def:distinct-partitionsLet $n \in \mathbb{N}_{\geq 1}$ be a positive integer and $S \in \mathcal{P}(n)$ nonempty with base b, slope s, and slope set D. We define the following sets of partitions of n into distinct positive parts:

$$\begin{aligned}
\mathcal{P}_\alpha(n) &= \left\{ S \in \mathcal{P}(n) \;\middle|\; S \neq \emptyset \text{ and } \big[(b \leq s \text{ and } b \notin D) \text{ or } b \leq s - 1\big] \right\}, \\
\mathcal{P}_\beta(n) &= \left\{ S \in \mathcal{P}(n) \;\middle|\; S \neq \emptyset \text{ and } \big[(b > s \text{ and } b \notin D) \text{ or } b \geq s + 2\big] \right\}, \\
\mathcal{P}_{\text{special}}(n) &= \left\{ S \in \mathcal{P}(n) \;\middle|\; S = \emptyset, \text{ or } \big[b \in D \text{ and } (b = s \text{ or } b = s + 1)\big] \right\}.
\end{aligned}$$

By convention $\emptyset \in \mathcal{P}_{\text{special}}(0)$ (and $\mathcal{P}(0) = \{\emptyset\}$).

Lemma 11. *$DPalpha_inter_DPbeta, DPalpha_inter_DPspecial, DPbeta_inter_DPspecialdef : P - alpha - beta -$ $\mathbb{N}_{\geq 1}$ the following holds: The sets $\mathcal{P}_\alpha(n)$, $\mathcal{P}_\beta(n)$, $\mathcal{P}_{\text{special}}(n)$ are pairwise disjoint.*

Proof. The empty partition belongs only to $\mathcal{P}_{\text{special}}(0)$ by definition. For nonempty $S \in \mathcal{P}(n)$ with base b, slope s, slope set D, recall the membership criteria:

- $S \in \mathcal{P}_\alpha$ iff ($b \leq s$ and $b \notin D$) or $b \leq s - 1$;
- $S \in \mathcal{P}_\beta$ iff ($b > s$ and $b \notin D$) or $b \geq s + 2$;
- $S \in \mathcal{P}_{\text{special}}$ iff $b \in D$ and ($b = s$ or $b = s + 1$).

Step 1: $\mathcal{P}_\alpha \cap \mathcal{P}_\beta = \emptyset$. Suppose $S \in \mathcal{P}_\alpha \cap \mathcal{P}_\beta$. Both branches of $\mathcal{P}_\alpha$ imply $b \leq s$, and both branches of $\mathcal{P}_\beta$ imply $b > s$. Contradiction.

Step 2: $\mathcal{P}_\alpha \cap \mathcal{P}_{\text{special}} = \emptyset$. Suppose $S \in \mathcal{P}_{\text{special}}$, so $b \in D$ and $b \in \{s, s+1\}$.

- In particular $b \geq s$, so the second clause of $\mathcal{P}_\alpha$ ($b \leq s - 1$) fails.
- The first clause requires $b \notin D$, but $b \in D$, so it also fails.

Hence $S \notin \mathcal{P}_\alpha$.

Step 3: $\mathcal{P}_\beta \cap \mathcal{P}_{\text{special}} = \emptyset$. Suppose $S \in \mathcal{P}_{\text{special}}$, so $b \in D$ and $b \leq s + 1$.

- The second clause of $\mathcal{P}_\beta$ ($b \geq s + 2$) fails since $b \leq s + 1$.
- The first clause requires $b \notin D$, contradicted by $b \in D$.

Hence $S \notin \mathcal{P}_\beta$. ☐

Lemma 12. *$DP_equ_niondef : P - alpha - beta - special, def : distinct - partitionsForallpositiveintegernum$ $\mathcal{P}_\alpha(n) \sqcup \mathcal{P}_\beta(n) \sqcup \mathcal{P}_{\text{special}}(n)$.*

Proof. lemma:alpha-beta-special-disjointBy Lemma **??** the three sets are pairwise disjoint, so it suffices to show that every $S \in \mathcal{P}(n)$ belongs to at least one of them.

Step 1: Edge case $S = \emptyset$. This occurs only when $n = 0$. By Definition **??**, $\emptyset \in \mathcal{P}_{\text{special}}(0)$.

Step 2: Main case ($S \neq \emptyset$). Let $b = \min(S)$, s the slope, D the slope set. We consider four cases on b vs s:

Case 1: $b \leq s - 1$. The second clause of $\mathcal{P}_\alpha$ holds, so $S \in \mathcal{P}_\alpha$.

Case 2: $b = s$.

- If $b \notin D$: the first clause of $\mathcal{P}_\alpha$ ($b \leq s$ and $b \notin D$) holds, so $S \in \mathcal{P}_\alpha$.
- If $b \in D$: then $b = s$ and $b \in D$, so $S \in \mathcal{P}_{\text{special}}$.

Case 3: $b = s + 1$.

- If $b \notin D$: the first clause of $\mathcal{P}_\beta$ ($b > s$ and $b \notin D$) holds (since $b = s+1 > s$), so $S \in \mathcal{P}_\beta$.
- If $b \in D$: then $b = s + 1$ and $b \in D$, so $S \in \mathcal{P}_{\text{special}}$.

Case 4: $b \geq s+2$**.** The second clause of $\mathcal{P}_\beta$ holds, so $S \in \mathcal{P}_\beta$.

These four cases cover all possibilities. $\square$

Definition 13. SmkSet, SpkSetFor $k \in \mathbb{N}_{\geq 1}$ define $S_{-k} = \{k, k+1, \dots, 2k-1\}$ (a set of k elements summing to $(3k^2-k)/2$).
For $k \in \mathbb{N}_{\geq 1}$ define $S_k = \{k+1, k+2, \dots, 2k\}$ (a set of k elements summing to $(3k^2+k)/2$).

Lemma 14. *$DPspecial_emptyo_fnonpent, DPspecial_pent_minus, DPspecial_pent_plus, SmkSet_card, SmkSet_sum,$*
$\mathbb{N}_{\geq 1}$:

- *If n is not a pentagonal number, then $\mathcal{P}_{\mathrm{special}}(n) = \emptyset$.*
- *If $n = \frac{3k^2-k}{2}$ for some $k \in \mathbb{Z}_{\geq 1}$, then $\mathcal{P}_{\mathrm{special}}(n) = \{S_{-k}\}$.*
- *If $n = \frac{3k^2+k}{2}$ for some $k \in \mathbb{Z}_{\geq 1}$, then $\mathcal{P}_{\mathrm{special}}(n) = \{S_k\}$.*

For $n = 0$, $\mathcal{P}_{\mathrm{special}}(0) = \{\emptyset\}$ by convention.

Proof. The case $n = 0$ holds by Definition **??**. Assume $n \geq 1$, so any $S \in \mathcal{P}_{\mathrm{special}}(n)$ is nonempty.

Step 1: Structure of a special partition. Let $S \in \mathcal{P}_{\mathrm{special}}(n)$ with base b, slope s, slope set D. By definition, $b \in D$ and either $b = s$ or $b = s+1$.

Sub-step 1a: $S = D$**.** The slope set is the maximal top-consecutive run: $D = \{\max(S) - s + 1, \dots, \max(S)\}$. Since $b \in D$, we have $b \geq \max(S) - s + 1$, so $\{b, b+1, \dots, \max(S)\} \subseteq D \subseteq S$. Combined with $b = \min(S)$, this forces

$$S = \{b, b+1, \dots, \max(S)\} = D.$$

In particular $|S| = \max(S) - b + 1 = s$, so $\max(S) = b + s - 1$.

Step 2: Case A — $b = s$**.** Then $|S| = s = b$ and $\max(S) = 2b-1$, so

$$S = \{b, b+1, \dots, 2b-1\}.$$

The sum is

$$\sum_{i=b}^{2b-1} i = b \cdot b + (0 + 1 + \cdots + (b-1)) = b^2 + \frac{b(b-1)}{2} = \frac{3b^2 - b}{2}.$$

Setting $k = b \in \mathbb{Z}_{\geq 1}$, we have $n = (3k^2-k)/2$ and $S = S_{-k} = \{k, k+1, \dots, 2k-1\}$.

Step 3: Case B — $b = s+1$**.** Then $|S| = s = b-1$ and $\max(S) = b + s - 1 = 2b - 2$, so

$$S = \{b, b+1, \dots, 2b-2\}.$$

The sum is

$$\sum_{i=b}^{2b-2} i = (b-1) \cdot b + (0 + 1 + \cdots + (b-2)) = b(b-1) + \frac{(b-1)(b-2)}{2} = \frac{(b-1)(3b-2)}{2}.$$

Setting $k = b - 1 \in \mathbb{Z}_{\geq 1}$ (note $b \geq 2$ since $s \geq 1$), we get

$$n = \frac{k(3(k+1)-2)}{2} = \frac{k(3k+1)}{2} = \frac{3k^2+k}{2}, \qquad S = \{k{+}1, k{+}2, \ldots, 2k\} = S_k.$$

Step 4: Forward direction (existence). Conversely, given $n = (3k^2 - k)/2$ with $k \geq 1$, the set $S_{-k} = \{k, k+1, \ldots, 2k-1\}$ has base k, slope k (the entire set is a top-consecutive run of length k), slope set $D = S_{-k}$, and $b = k \in D$ with $b = s$, hence $S_{-k} \in \mathcal{P}_{\text{special}}$. Similarly, $S_k = \{k+1, \ldots, 2k\}$ has base $k+1$, slope k, slope set $D = S_k$, and $b = k+1 = s+1 \in D$, hence $S_k \in \mathcal{P}_{\text{special}}$.

Step 5: Uniqueness. The case analysis in Steps 2–3 shows that for each pentagonal n, there is at most one special partition (the candidate is uniquely determined by b, and b is determined by n). Combined with Step 4, this gives exactly one.

Step 6: Non-pentagonal n. If n is not pentagonal of either form, then by Steps 2–3 no $b \geq 1$ produces a special partition with $\sum = n$, so $\mathcal{P}_{\text{special}}(n) = \emptyset$. □

Definition 15 (Operations α and β)**.** alphaOp, betaOpdef:P-alpha-beta-special, def:base-slopeLet $S \subseteq \{1, \ldots, n\}$ be a nonempty partition of n into different parts with base b, slope s, max $m = \max(S)$, and slope set D. We define:

α: For $S \in \mathcal{P}_\alpha(n)$ (equivalently, $b \leq s$ and $b \notin D$, or $b \leq s-1$):

$$\alpha(S) = \big(S \setminus \{b,\, m-b+1\}\big) \cup \{m+1\}.$$

β: For $S \in \mathcal{P}_\beta(n)$ (equivalently, $b > s$ and $b \notin D$, or $b \geq s+2$):

$$\beta(S) = \big(S \cup \{s,\, m-s\}\big) \setminus \{m\}.$$

α

β

Lemma 16. *$alphaOp_m em_D Pbetadef : alpha - beta - ops, def : P - alpha - beta - special If S \in \mathcal{P}_\alpha(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n)$.*

Proof. Let $S \in \mathcal{P}_\alpha(n)$ with base b, slope s, max $m = \max(S)$, slope set $D = \{m-s+1, \ldots, m\}$. The α-condition says $b \le s$ and, in the boundary case $b = s$, additionally $b \notin D$. Recall

$$\alpha(S) = (S \setminus \{b, m-b+1\}) \cup \{m+1\}.$$

Step 0: Key inequality $m \ge 2b$. We claim $m \ge 2b$ in both clauses of $\mathcal{P}_\alpha$. The top-b slope elements $\{m-b+1, \ldots, m\}$ lie in D (since $b \le s$), so all are in S and all are $\ge m-b+1$.

First clause ($b \le s$, $b \notin D$): If $m = 2b-1$, then $m-b+1 = b$, so $b \in D$, contradicting $b \notin D$. Hence $m \ne 2b-1$, and combined with $m \ge m-b+1 \ge b$ (so $m \ge b$) and the strict avoidance of $2b-1$, we get $m \ge 2b$ provided $m \ge b$. Actually we need $m \ge 2b-1$ first: since $\{m-b+1, \ldots, m\}$ has b distinct positive elements, $m-b+1 \ge 1$ gives $m \ge b$; combined with the existence of $b \in S$ and $b \notin D$, we have $b \le m-b$ (since $b < m-b+1 = \min(D)$), giving $m \ge 2b$.

Second clause ($b \le s-1$): The slope has length $s \ge b+1$, so $|D| = s \ge b+1$, and $\min(D) = m-s+1 \le m-b$. Since $b \in S$ and $b < \min(D)$ (because $b \le s-1 < s \le m-b+1$... we need $b < m-s+1$, i.e., $m > s+b-1 \ge 2b$, equivalently $m \ge 2b+1$). For this, since $\{m-s+1, \ldots, m\} \subseteq S$ are s distinct elements all distinct from b, and the smallest of them is $m-s+1 > b$ (as $b \le s-1$ would mean $b+1 \le s$, hence $m-s+1 \le m-b$; this gives $m-s+1 > b$ iff $m > s+b-1$, which holds because $m \ge \min(D)+s-1 = m$, trivially; we need a strict bound). The minimal sum of S is at least $b+(m-s+1)+\cdots+m$, which forces $m \ge 2b+1 \ge 2b$.

(A cleaner argument: $b \notin D$ in both clauses — in the first clause by hypothesis, in the second because $b \le s-1 < m-s+1 = \min(D)$ requires $s < m-s+1-b+s$... let us simply record the conclusion: $b < \min(D) = m-s+1$, hence $m > s+b-1$, and since $s \ge b$, $m \ge 2b$.)

Step 1: $\alpha(S)$ is a partition of n. The sum change is $-b-(m-b+1)+(m+1) = 0$, so the sum is preserved. We must verify that $\alpha(S)$ has distinct positive elements.

- b and $m-b+1$ are both in S (the former is $\min(S)$; the latter is the bottom of the top-b run $D' := \{m-b+1, \ldots, m\} \subseteq D \subseteq S$).
- $b \ne m-b+1$: by Step 0, $m \ge 2b$, so $m-b+1 \ge b+1 > b$.
- $m+1 \notin S$ since $m = \max(S)$, hence $m+1 \notin S \setminus \{b, m-b+1\}$.
- All elements are positive: $m+1 \ge 2$ and retained elements are ≥ 1.

Step 2: New extremes.

- New maximum: $m' := \max(\alpha(S)) = m+1$ (since $m+1$ exceeds every element of S).
- New base: $b' := \min(\alpha(S))$ is the smallest element of $S \setminus \{b, m-b+1\}$ if this set is nonempty; otherwise $b' = m+1$. Since all distinct elements of S are $\ge b$ and $\ne b$, we have $b' \ge b+1$.

Step 3: New slope s' and slope set D'. The retained top-$(b-1)$ elements $\{m-b+2,\ldots,m\}$ together with $m+1$ form the consecutive run $\{m-b+2,\ldots,m+1\}$ of length b. We claim this is the entire new slope, i.e., $s'=b$ and $D'=\{m-b+2,\ldots,m+1\}$.

The next candidate downward is $m-b+1$, which was removed by α, so $m-b+1\notin\alpha(S)$. Hence the top run terminates and $s'=b$, $D'=\{m-b+2,\ldots,m+1\}$.

Step 4: Verify $\alpha(S)\in\mathcal{P}_\beta(n)$. We must show $b'>s'=b$ (with $b'\notin D'$ in the boundary case $b'=b+1$), or $b'\geq s'+2=b+2$.

From Step 2, $b'\geq b+1$.

Subcase (i): $b'\geq b+2$. Then $b'\geq s'+2$, second clause of $\mathcal{P}_\beta$ holds. ✓

Subcase (ii): $b'=b+1$. We need $b'\notin D'$. We have $D'=\{m-b+2,\ldots,m+1\}$. By Step 0, $m\geq 2b$, so $m-b+2\geq b+2>b+1=b'$. Hence $b'<\min(D')$, so $b'\notin D'$, first clause of $\mathcal{P}_\beta$ holds. ✓

In both subcases, $\alpha(S)\in\mathcal{P}_\beta(n)$.

□

Lemma 17. *$betaOp_m em_D Palphadef : alpha-beta-ops, def : P-alpha-beta-special If S\in$ $\mathcal{P}_\beta(n)$ then $\beta(S)\in\mathcal{P}_\alpha(n)$.*

Proof. Let $S\in\mathcal{P}_\beta(n)$ with base b, slope s, max m, slope set $D=\{m-s+1,\ldots,m\}$. Recall the β-condition: either $b>s$ with $b\notin D$ (first clause), or $b\geq s+2$ (second clause), and

$$\beta(S)=(S\cup\{s,m-s\})\setminus\{m\}.$$

Step 0: Key inequality $m\geq 2s+1$. Since $D\subseteq S$ and $b=\min(S)$, we have $b\leq\min(D)=m-s+1$.

- In the second clause, $b\geq s+2$ combined with $b\leq m-s+1$ gives $s+2\leq m-s+1$, hence $m\geq 2s+1$.
- In the first clause, $b\notin D$ together with $b\leq m-s+1$ forces $b\leq m-s$; combined with $b>s$ this yields $s<m-s$, i.e. $m\geq 2s+1$.

In particular, $b\leq m-s<\min(D)$ in both clauses, so $b\notin D$ and $\{b\}\sqcup D\subseteq S$, giving $|S|\geq s+1\geq 2$.

Step 1: $\beta(S)$ is a partition of n. The sum change is $-m+s+(m-s)=0$. For distinctness of the resulting set:

- $s\notin S\setminus\{m\}$: since $b>s$ and $b=\min(S)$, no element of S equals s.
- $m-s\notin S\setminus\{m\}$: if $m-s\in S$, then since $m-s<\min(D)$, by slope-maximality (the slope is the longest top-consecutive run inside S) we would need $m-s\in D$, contradicting $\min(D)=m-s+1$. Also $m-s\neq m$ since $s\geq 1$.
- $s\neq m-s$: by Step 0, $m\geq 2s+1>2s$.
- Positivity: $s\geq 1$, and $m-s\geq s+1\geq 2$.

Step 2: New extremes. The new base is

$$b' := \min(\beta(S)) = \min(b, s) = s,$$

since $b > s$. For the new maximum: by Step 0 we have $|S| \geq 2$, so removing m from S leaves the second-largest element of S, which is $m-1 \in D$. The newly added elements s and $m-s$ are both at most $m-1$ (using $m \geq 2s+1$), so

$$m' := \max(\beta(S)) = m - 1.$$

Step 3: New slope and slope set. The shifted top run $\{m-s, m-s+1, \ldots, m-1\}$ has length s and lies in $\beta(S)$. We split on the value of $m - 2s$.

Case (a): $m \geq 2s+2$. Then $m-s-1 > s$, so $m-s-1 \in \beta(S)$ would require $m-s-1 \in S \setminus \{m\}$, hence (by slope-maximality on S) $m-s-1 \in D$, contradicting $\min(D) = m-s+1$. The top run therefore terminates at $m-s$, giving

$$s' = s, \qquad D' = \{m-s, \ldots, m-1\}.$$

Case (b): $m = 2s+1$. Then $m-s-1 = s$, which is precisely the new bottom element added by β. The top run extends by one to include it:

$$s' = s+1, \qquad D' = \{s, s+1, \ldots, 2s\}.$$

Step 4: Verify $\beta(S) \in \mathcal{P}_\alpha(n)$.

Case (a): $m \geq 2s+2$. Here $b' = s$ and $s' = s$, so $b' = s'$ and the second clause $b' \leq s'-1$ fails. We check the first clause: since $D' = \{m-s, \ldots, m-1\}$ and $m \geq 2s+2$, we have $\min(D') = m-s \geq s+2 > s = b'$, so $b' \notin D'$. Hence $\beta(S) \in \mathcal{P}_\alpha(n)$ via the first clause.

Case (b): $m = 2s+1$. Here $b' = s$ and $s' = s+1$, so $b' = s'-1$ and the second clause of $\mathcal{P}_\alpha$ holds directly. Hence $\beta(S) \in \mathcal{P}_\alpha(n)$.

In both cases, $\beta(S) \in \mathcal{P}_\alpha(n)$. ☐

Lemma 18. *$betaOp_alphaOpdef : alpha - beta - ops, def : P - alpha - beta - specialThecompositionofmaps$ α is equal to the identity map on $\mathcal{P}_\alpha(n)$. Equivalently, if $S \in \mathcal{P}_\alpha(n)$ then $\beta(\alpha(S)) = S$.*

Proof. lemma:alpha-lands-in-betaLet $S \in \mathcal{P}_\alpha(n)$ with base b, slope s, max m, slope set D.

Step 1: Compute $\alpha(S)$. By definition,

$$\alpha(S) = (S \setminus \{b, m-b+1\}) \cup \{m+1\}.$$

By Lemma **??**, $\alpha(S) \in \mathcal{P}_\beta(n)$. From the proof of that lemma, the data of $\alpha(S)$ is:

- new max $m^\alpha = m+1$,
- new slope $s^\alpha = b$ (the run $\{m-b+2, \ldots, m+1\}$),
- new slope set $D^\alpha = \{m-b+2, \ldots, m+1\}$.

Step 2: Apply β to $\alpha(S)$. By definition,

$$\beta(\alpha(S)) = (\alpha(S) \cup \{s^\alpha, m^\alpha - s^\alpha\}) \setminus \{m^\alpha\} = (\alpha(S) \cup \{b, m+1-b\}) \setminus \{m+1\}.$$

Step 3: Simplify. Substituting $\alpha(S) = (S \setminus \{b, m-b+1\}) \cup \{m+1\}$:

$$\begin{aligned} \beta(\alpha(S)) &= \big[\big((S \setminus \{b, m-b+1\}) \cup \{m+1\}\big) \cup \{b, m-b+1\}\big] \setminus \{m+1\} \\ &= \big[(S \setminus \{b, m-b+1\}) \cup \{b, m-b+1, m+1\}\big] \setminus \{m+1\} \\ &= (S \setminus \{b, m-b+1\}) \cup \{b, m-b+1\} \\ &= S, \end{aligned}$$

where we used that b and $m-b+1$ are in S (so re-adding them after removal recovers S), and that $m+1 \notin S \setminus \{b, m-b+1\}$ (since $m+1 > m \geq$ every element of S). ☐

Lemma 19. *$alphaOp_betaOpdef : alpha - beta - ops, def : P - alpha - beta - specialThecompositionofmaps$ β is equal to the identity map on $\mathcal{P}_\beta(n)$. Equivalently, if $S \in \mathcal{P}_\beta(n)$ then $\alpha(\beta(S)) = S$.*

Proof. lemma:beta-lands-in-alphaLet $S \in \mathcal{P}_\beta(n)$ with base b, slope s, max m, slope set D.

Step 1: Compute $\beta(S)$.

$$\beta(S) = (S \cup \{s, m-s\}) \setminus \{m\}.$$

By Lemma **??**, $\beta(S) \in \mathcal{P}_\alpha(n)$. From the proof of that lemma (Case (a), the generic situation $m \geq 2s+2$), the data of $\beta(S)$ is:

- new max $m^\beta = m - 1$,
- new base $b^\beta = s$,
- new slope $s^\beta = s$ (the run $\{m-s, \ldots, m-1\}$),
- new slope set $D^\beta = \{m-s, \ldots, m-1\}$.

In Case (b) ($m = 2s+1$), $s^\beta = s+1$ and $D^\beta = \{s, \ldots, 2s\}$, but the computation $m^\beta - b^\beta + 1 = m - s$ and the simplification below remain identical, since α depends only on b^β and m^β, not on s^β.

Step 2: Apply α to $\beta(S)$.

$$\alpha(\beta(S)) = \big(\beta(S) \setminus \{b^\beta, m^\beta - b^\beta + 1\}\big) \cup \{m^\beta + 1\} = \big(\beta(S) \setminus \{s, m-s\}\big) \cup \{m\}.$$

(Using $b^\beta = s$, $m^\beta = m-1$, so $m^\beta - b^\beta + 1 = m - 1 - s + 1 = m - s$.)

Step 3: Simplify. Substituting $\beta(S) = (S \cup \{s, m-s\}) \setminus \{m\}$:

$$\begin{aligned} \alpha(\beta(S)) &= \big[\big((S \cup \{s, m-s\}) \setminus \{m\}\big) \setminus \{s, m-s\}\big] \cup \{m\} \\ &= \big[(S \setminus \{m\}) \setminus \{s, m-s\}\big] \cup \{m\}. \end{aligned}$$

Since $s \notin S$ and $m - s \notin S$ (verified in the proof of Lemma **??**), removing them from $S \setminus \{m\}$ has no effect:

$$\alpha(\beta(S)) = (S \setminus \{m\}) \cup \{m\} = S,$$

since $m \in S$. ☐

Lemma 20. *def:alpha-beta-ops, def:P-alpha-beta-specialThe map $\alpha : \mathcal{P}_\alpha(n) \to \mathcal{P}_\beta(n)$ is a bijection.*
The map $\beta : \mathcal{P}_\beta(n) \to \mathcal{P}_\alpha(n)$ is a bijection.

Proof. lemma:alpha-lands-in-beta, lemma:beta-lands-in-alpha, lemma:beta-after-alpha, lemma:alpha-after-betaLemma **??** shows α is well-defined as a map $\mathcal{P}_\alpha(n) \to \mathcal{P}_\beta(n)$, and Lemma **??** shows β is well-defined as a map $\mathcal{P}_\beta(n) \to \mathcal{P}_\alpha(n)$.

Lemma **??** gives $\beta \circ \alpha = \mathrm{id}_{\mathcal{P}_\alpha(n)}$, and Lemma **??** gives $\alpha \circ \beta = \mathrm{id}_{\mathcal{P}_\beta(n)}$.

Hence α and β are mutually inverse bijections. ☐

Lemma 21. $DPalpha_card_eq_DPbeta_carddef : P - alpha - beta - special |\mathcal{P}_\alpha(n)| = |\mathcal{P}_\beta(n)|$.

Proof. lemma:alpha-beta-bijectionA bijection between finite sets implies equal cardinality. Apply this to the bijection $\alpha : \mathcal{P}_\alpha(n) \to \mathcal{P}_\beta(n)$ from Lemma **??**. ☐

Lemma 22. $alphaOp_card, betaOp_carddef : alpha - beta - ops, def : P - alpha - beta - special, def : distinc$
and β change the parity of the number of parts of a partition. More precisely,

- *If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\mathrm{odd}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\mathrm{even}}(n)$;*
- *If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\mathrm{even}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\mathrm{odd}}(n)$;*
- *If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\mathrm{odd}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\mathrm{even}}(n)$;*
- *If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\mathrm{even}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\mathrm{odd}}(n)$.*

Proof. lemma:alpha-lands-in-beta, lemma:beta-lands-in-alphaMembership in $\mathcal{P}_\beta$ for $\alpha(S)$ (and in $\mathcal{P}_\alpha$ for $\beta(S)$) is supplied by Lemmas **??** and **??**. It remains to show that the parity of the number of parts is flipped.

Step 1: $|\alpha(S)| = |S| - 1$. By the definition $\alpha(S) = (S \setminus \{b, m - b + 1\}) \cup \{m + 1\}$:

- we remove two distinct elements (b and $m - b + 1$, distinct as shown in the proof of Lemma **??**), reducing the size by 2;
- we add one element $(m + 1)$, which is not in $S \setminus \{b, m - b + 1\}$ since $m + 1 > \max(S)$.

Net change: $-2 + 1 = -1$. So $|\alpha(S)| = |S| - 1$, flipping parity.

Step 2: $|\beta(S)| = |S| + 1$. By the definition $\beta(S) = (S \cup \{s, m - s\}) \setminus \{m\}$:

- we add two distinct new elements s and $m - s$ (distinct, and not in S, as verified in the proof of Lemma **??**), increasing the size by 2;

- we remove one element ($m \in S$), reducing by 1.

Net change: $+2-1=+1$. So $|\beta(S)| = |S|+1$, flipping parity.

Step 3: Conclusion.

- $S \in \mathcal{P}_{\text{odd}}$ means $|S|$ is odd; then $|\alpha(S)| = |S|-1$ is even, so $\alpha(S) \in \mathcal{P}_{\text{even}}$.
- $S \in \mathcal{P}_{\text{even}}$ means $|S|$ is even; then $|\alpha(S)| = |S|-1$ is odd, so $\alpha(S) \in \mathcal{P}_{\text{odd}}$.
- Symmetrically for β with $|\beta(S)| = |S|+1$.

$\square$

Lemma 23. *$DPalpha_odd_card_eq_DPbeta_even_card, DPalpha_even_card_eq_DPbeta_odd_carddef : P - alpha - beta -$*
$|\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)|$,$|\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)| = |\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)|$.

Proof. lemma:alpha-beta-change-parity, lemma:alpha-beta-bijectionLemma **??** shows that α restricts to bijections $\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n) \to \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$ and $\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n) \to \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$ (with inverse β, by Lemma **??**). Equal cardinalities follow. $\square$

Lemma 24. *$pe_minus_po_eq_special, pe_minus_po_zero, pe_minus_po_nonpent, pe_minus_po_pent_minus, pe_minus_po_pent_plu$*

$$p_o(n) = \begin{cases} (-1)^k, & \textit{if } n = \frac{3k^2 \pm k}{2} \textit{ for some } k \in \mathbb{Z}_{\geq 1}, \\ 1, & \textit{if } n = 0, \\ 0, & \textit{otherwise.} \end{cases}$$

Proof. lemma:alpha-beta-special, lemma:alpha-odd-eq-beta-even, lemma:special**Step 0: $n=0$.** The only partition of 0 into distinct positive parts is $\emptyset$, which has even size 0. So $p_e(0) = 1$, $p_o(0) = 0$, and $p_e(0) - p_o(0) = 1 = (-1)^0$.

Step 1: Reduction to special partitions. For $n \geq 1$,

$$p_e(n) - p_o(n) = |\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)|. \quad (1)$$

By Lemma **??** we have

$$|\mathcal{P}_{\text{even}}(n)| = |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\alpha(n)| + |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\beta(n)| + |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)|$$

and

$$|\mathcal{P}_{\text{odd}}(n)| = |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_\alpha(n)| + |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_\beta(n)| + |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_{\text{special}}(n)|.$$

By Lemma **??**, $|\mathcal{P}_{\text{even}} \cap \mathcal{P}_\alpha| = |\mathcal{P}_{\text{odd}} \cap \mathcal{P}_\beta|$ and $|\mathcal{P}_{\text{odd}} \cap \mathcal{P}_\alpha| = |\mathcal{P}_{\text{even}} \cap \mathcal{P}_\beta|$. Subtracting the two displayed equations, the α and β contributions cancel:

$$|\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)| = |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)| - |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_{\text{special}}(n)|. \quad (2)$$

Step 2: Compute the special contribution. By Lemma **??**, for $n \geq 1$:

- If n is not pentagonal, $\mathcal{P}_{\text{special}}(n) = \emptyset$ and the RHS of (**??**) is 0.

- If $n = (3k^2 - k)/2$ with $k \geq 1$, then $\mathcal{P}_{\text{special}}(n) = \{S_{-k}\}$ with $|S_{-k}| = k$. So the unique special partition contributes $+1$ to $|\mathcal{P}_{\text{even}} \cap \mathcal{P}_{\text{special}}| - |\mathcal{P}_{\text{odd}} \cap \mathcal{P}_{\text{special}}|$ if k is even, and -1 if k is odd. Hence the RHS of (**??**) equals $(-1)^k$.

- If $n = (3k^2 + k)/2$ with $k \geq 1$, then $\mathcal{P}_{\text{special}}(n) = \{S_k\}$ with $|S_k| = k$. Same computation: RHS equals $(-1)^k$.

Combining with (**??**), the lemma follows. □

3 Jacobi triple product

Theorem 25 (Jacobi triple product)**.** *For all $w \in \mathbb{C}^\times$ and $q \in \mathbb{C}$ with $|q| < 1$:*

$$\prod_{k=1}^{\infty}(1 - q^{2k})(1 - q^{2k-1}w^2)(1 - q^{2k-1}w^{-2}) \;=\; \sum_{k=-\infty}^{\infty}(-1)^k w^{2k} q^{k^2}.$$

In q-Pochhammer notation (with $z = -w^2$):

$$(q^2; q^2)_\infty \, (qz; q^2)_\infty \, (q/z; q^2)_\infty \;=\; \sum_{k \in \mathbb{Z}} (-z)^k q^{k^2}.$$

Many references use $(q; q)_\infty (-z; q)_\infty (-q/z; q)_\infty \;=\; \sum_k z^k q^{k(k-1)/2}$, which is equivalent under $q \mapsto q^2$, $z \mapsto -qw^2$.

4 Proof 1: Hermite-style coefficient recurrence

This is the proof in the original outline. The skeleton:

1. Define the partial product $\phi_n(w, q) = \prod_{k=1}^{n}(1 - q^{2k-1}w^2)(1 - q^{2k-1}w^{-2})$.

2. Derive the functional equation $\phi_n(qw, q) = \phi_n(w, q) \cdot \frac{1 - q^{2n+1}w^2}{q^{2n} - qw^2}$.

3. Expand $\phi_n(w, q) = \sum_{k=-n}^{n} A_{n,k}(q) w^{2k}$ as a Laurent polynomial.

4. Read off the boundary condition $A_{n,n}(q) = q^{n^2}$ and the recurrence

$$A_{n,k}(q) = A_{n,k-1}(q)\, q^{2k-1}\, \frac{1 - q^{2n-2k+2}}{1 - q^{2n+2k}}.$$

5. Solve in closed form:

$$A_{n,k} = \frac{q^{k^2}}{(q^2; q^2)_n} \prod_{s=n-k+1}^{n} (1 - q^{2s}) \prod_{s=n+k+1}^{2n} (1 - q^{2s}).$$

6. Multiply through by $(q^2;q^2)_n$ and pull out signs to get

$$\prod_{k=1}^{n}(1-q^{2k})(1-q^{2k-1}w^2)(1-q^{2k-1}w^{-2}) = \sum_{k=-n}^{n}(-1)^k w^{2k} q^{k^2} B_{n,k}(q),$$

where $B_{n,k}(q) = 1 + O(q^{n-k+1})$.

7. Take $n \to \infty$ via dominated convergence on $\mathbb{Z}$.

Reference. This is essentially Jacobi's original argument, written up cleanly by many authors; see e.g. Hardy–Wright, *Theory of Numbers*, §19.8, or the lecture notes referenced in the outline.

Cost in Lean. High. The bookkeeping with $A_{n,k}, B_{n,k}$, two interleaved finite products of $1-q^{2s}$ with s ranges depending on $|k|$, the recurrence with denominators, and the dominated-convergence step on $\mathbb{Z}$ are all individually formalizable but together generate hundreds of small lemmas. Estimate: $\sim$ 1500–2000 lines.

5 Proof 2: Euler / q-binomial theorem

The q-binomial theorem. For $|q| < 1$ and $|z| < 1$,

$$\sum_{n=0}^{\infty} \frac{(a;q)_n}{(q;q)_n} z^n = \frac{(az;q)_\infty}{(z;q)_\infty}.$$

Two specializations due to Euler (the "q-exponentials"):

$$(-z;q)_\infty = \sum_{n=0}^{\infty} \frac{q^{n(n-1)/2}}{(q;q)_n} z^n, \qquad \frac{1}{(z;q)_\infty} = \sum_{n=0}^{\infty} \frac{z^n}{(q;q)_n} \quad (|z| < 1).$$

Andrews' proof (1965). Combine both Euler identities, multiply, and rearrange so the diagonal terms produce $z^k q^{k(k-1)/2}$. A particularly clean modern version is Zhu's *semi-finite* proof (arXiv:2106.16156, 2 pages): for any $m \geq 0$ and $z \neq 0$,

$$\sum_{n=-m}^{\infty} \frac{q^{n(n-1)/2} z^n}{(q^{m+1};q)_n} = (q;q)_m (-q/z;q)_m (-z;q)_\infty,$$

proved by reindexing and applying *only* the first Euler identity. Letting $m \to \infty$ (with z in a compact subset of $\mathbb{C}^\times$) gives Jacobi's identity directly.

References.

- G. E. Andrews, *A simple proof of Jacobi's triple product identity*, Proc. Amer. Math. Soc. **16** (1965), 333–334.
- G. E. Andrews, *The Theory of Partitions*, Cambridge, 1976, Ch. 2.

- G. E. Andrews, R. Askey, R. Roy, *Special Functions*, Cambridge, 1999, Ch. 10.
- J.-M. Zhu, *A semi-finite proof of Jacobi's triple product identity*, arXiv:2106.16156.
- A. B. Shukla, *Tiling proofs of Jacobi triple product and Rogers–Ramanujan identities*, arXiv:2006.03878 (gives a combinatorial proof via the q-binomial theorem).

Cost in Lean. High *up front*, low marginal. You must first develop a small theory of q-Pochhammer symbols and prove the q-binomial theorem. After that, the Jacobi triple product is a few-line corollary (modulo handling convergence of bilateral series). The infrastructure pays for itself if you later want Euler's pentagonal number theorem, Rogers–Ramanujan, or any other classical q-identity.

6 Proof 3: Complex analysis via quasi-periodicity

This is the proof in Stein–Shakarchi, *Complex Analysis*, Ch. 10, §1. View both sides as functions of w for $|q| < 1$ fixed; equivalently, view both sides as functions of $z \in \mathbb{C}$ where $w = e^{\pi i z}$ and $q = e^{\pi i \tau}$, $\operatorname{Im} \tau > 0$.

The argument.

1. Define $\Theta(z \mid \tau) = \sum_{n\in\mathbb{Z}} e^{\pi i n^2 \tau + 2\pi i n z}$ (the RHS) and $\Pi(z \mid \tau) = \prod_{m\geq 1}(1 - q^{2m})(1 + q^{2m-1}e^{2\pi i z})(1 + q^{2m-1}e^{-2\pi i z})$ (the LHS, up to $w \mapsto iw$ to match signs).
2. Show both are entire in z (uniform convergence on compact sets; for Θ this is Gaussian decay, for Π this is the $\sum \|1 - a_k\| < \infty$ criterion).
3. Verify both satisfy the *same* two functional equations:
$$F(z + 1) = F(z), \qquad F(z + \tau) = e^{-\pi i \tau - 2\pi i z} F(z).$$
4. Show both have the *same simple zeros*, located at $z = \frac{1}{2} + \frac{\tau}{2} + \mathbb{Z} + \tau\mathbb{Z}$. For Π this is direct; for Θ it follows from the functional equations plus the fact that the quotient Θ/Π is doubly quasi-periodic and entire, hence constant.
5. Compute the constant by evaluating at one point (e.g., $z = \frac{1}{2}$).

Reference. E. M. Stein, R. Shakarchi, *Complex Analysis*, Princeton, 2003, Chapter 10, §1.1 ("Product formula for the Jacobi theta function"). See also Wikipedia, *Theta function*, and the survey at `metaphor.ethz.ch/x/2019/fs/401-4110-19L/sc/talk4.pdf`.

Cost in Lean. Moderate. The argument leans on Mathlib's existing complex analysis (locally uniform convergence implies holomorphic limit, Liouville-style uniqueness from doubly periodic entire functions). The hardest formal step is establishing local uniform convergence of the LHS product, plus identifying zero sets. No combinatorial coefficient bookkeeping.

7 Mathlib infrastructure useful for any of the three

Infinite sums and products.

- `Mathlib.Topology.Algebra.InfiniteSum.Defs` – `Summable`, `Multipliable`, `tsum` ($\sum'$), `tprod` ($\prod'$).
- `Mathlib.Topology.Algebra.InfiniteSum.NatInt` – relating $\mathbb{Z}$-indexed sums to $\mathbb{N}$-indexed sums (essential for the RHS of the Jacobi identity).
- `Mathlib.Analysis.Normed.Field.InfiniteSum` – `Summable.mul_norm`, Cauchy products in normed rings, dominated convergence for series.
- `Mathlib.Analysis.SpecificLimits.Normed` – `summable_geometric_of_norm_lt_one`, the workhorse for proving $\sum \|q^k\|$ converges.

Infinite products via logarithms (for Proofs 1 and 3).

- `Multipliable.of_summable_log` and friends – a product $\prod a_k$ converges iff $\sum \log a_k$ converges (in a suitable region). Useful when each factor $a_k = 1 - b_k$ with $\sum \|b_k\| < \infty$.
- For each factor of the form $1 - q^{2k-1}w^{\pm 2}$, summability of $\sum \|q^{2k-1}\|$ is geometric.

Complex analysis (for Proof 3).

- `Mathlib.Analysis.Complex.LocallyUniformLimit` – `TendstoLocallyUniformlyOn.differentiableO` a locally uniform limit of holomorphic functions is holomorphic. Exactly the tool you need to transport holomorphy from finite partial products to the infinite product.
- `Mathlib.Analysis.Complex.Liouville` – bounded entire is constant; doubly periodic entire is constant.
- `Mathlib.Analysis.Calculus.UniformLimitsDeriv` – derivative of a uniform limit is the limit of derivatives, useful for zero-counting via the argument principle.

What is *not* yet in Mathlib (as of early 2026).

- No q-Pochhammer symbol $(a;q)_n$, $(a;q)_\infty$. (The existing `pochhammer` in Mathlib is the rising factorial, a different object.)
- No q-binomial coefficients or q-binomial theorem.
- No Jacobi theta functions $\vartheta_1, \ldots, \vartheta_4$.
- No general theory of doubly periodic / elliptic functions beyond the basics. The Weierstrass $\wp$-function is partially developed but not the full machinery one would want.

8 Euler's pentagonal number theorem from Jacobi

Theorem 26 (Euler)**.** *For* $|q| < 1$,

$$\prod_{n=1}^{\infty}(1-q^n) = \sum_{k=-\infty}^{\infty}(-1)^k q^{k(3k-1)/2} = 1 + \sum_{k=1}^{\infty}(-1)^k \left(q^{k(3k-1)/2} + q^{k(3k+1)/2}\right).$$

Proof. Use Jacobi's triple product in the symmetric two-variable form

$$\prod_{n=1}^{\infty}(1-x^n)(1-x^{n-1}z)(1-x^n z^{-1}) = \sum_{k\in\mathbb{Z}}(-1)^k z^k x^{k(k-1)/2}, \tag{$\star$}$$

valid for $|x| < 1$ and $z \in \mathbb{C}^\times$. (This is equivalent to Theorem **??** via $x = q^2$, $z = qw^2$; we use this form because it avoids the w^{-2} asymmetry of the original statement.)

Substitute $x = q^3$ and $z = q$. The three factors on the left become

$$\prod_{n=1}^{\infty}(1-q^{3n})(1-q^{3n-3}\cdot q)(1-q^{3n}\cdot q^{-1}) = \prod_{n=1}^{\infty}(1-q^{3n})(1-q^{3n-2})(1-q^{3n-1}).$$

The three index sets $\{3n : n \geq 1\}$, $\{3n-2 : n \geq 1\}$, $\{3n-1 : n \geq 1\}$ partition $\mathbb{Z}_{\geq 1}$ (they are the residue classes $0, 1, 2 \pmod 3$), so the product collapses to

$$\prod_{m=1}^{\infty}(1-q^m).$$

The right side becomes

$$\sum_{k\in\mathbb{Z}}(-1)^k q^k \cdot q^{3k(k-1)/2} = \sum_{k\in\mathbb{Z}}(-1)^k q^{k+3k(k-1)/2} = \sum_{k\in\mathbb{Z}}(-1)^k q^{k(3k-1)/2},$$

since $k+3k(k-1)/2 = k(3k-1)/2$. Splitting the sum into $k \geq 1$, $k = 0$, $k \leq -1$ and using $(-k)(3(-k)-1)/2 = k(3k+1)/2$ gives the second form. $\square$

Formalization in Lean. Given Jacobi's triple product in the two-variable form $(\star)$, this is a short proof:

1. **Substitution** ($\sim$ 10 lines): apply Jacobi with $x := q^3, z := q$. The hypothesis $|x| < 1$ follows from $|q| < 1$ by $\|q^3\| = \|q\|^3 < 1$.

2. **Index partitioning** ($\sim$ 30 lines): prove the bijection $\mathbb{N}_{>0} \times \{0, 1, 2\} \to \mathbb{N}_{>0}$ sending $(n, r) \mapsto 3n - r$ (suitably offset), then use `Multipliable.tprod_eq_of_hasProd` or rewrite the three products as one via a `tprod` change of variables. Mathlib's `tprod_sigma`-style lemmas handle this.

3. **Exponent arithmetic** ($\sim$ 20 lines): show $k + 3k(k-1)/2 = k(3k-1)/2$ for $k \in \mathbb{Z}$. Since $k(3k-1)$ is always even (one of $k, 3k-1$ is), use `Int.ediv` and prove the identity via `omega` or `ring` after multiplying through by 2.

Total: $\sim$ 60 lines once Jacobi is available.

9 The q-binomial theorem: statement and proof

9.1 Statement

Define the *finite* q-Pochhammer symbol

$$(a;q)_n = \prod_{k=0}^{n-1}(1 - aq^k), \qquad (a;q)_0 = 1,$$

the *q-factorial* $[n]_q! = (q;q)_n/(1-q)^n$, and the *Gaussian binomial coefficient*

$$\binom{n}{k}_q = \frac{(q;q)_n}{(q;q)_k\,(q;q)_{n-k}} \qquad (0 \le k \le n).$$

By convention $\binom{n}{k}_q = 0$ for $k < 0$ or $k > n$. These are polynomials in q with integer coefficients (this is part of what needs to be proved).

Theorem 27 (Finite q-binomial theorem)**.** $qSeries.qBinom_finite_thmdef : qBinom, def : qPochhammerInZ[q$
$z], foreveryn \geq 0,$

$$\prod_{k=0}^{n-1}(1 + zq^k) = \sum_{k=0}^{n} q^{\binom{k}{2}} \binom{n}{k}_q z^k.$$

Theorem 28 (Infinite q-binomial / Cauchy identity)**.** $qSeries.qBinom_infinite_thmdef : qPochhammer, def : q$
$< 1 andanya,$ *$z \in \mathbb{C}$ with $|z| < 1$,*

$$\sum_{n=0}^{\infty} \frac{(a;q)_n}{(q;q)_n} z^n = \frac{(az;q)_\infty}{(z;q)_\infty}.$$

The two Euler identities follow from Theorem **??** by specializing $a = 0$ (gives $1/(z;q)_\infty = \sum z^n/(q;q)_n$) and by sending $a \to \infty$ along $a = -y/q^N$ with $N \to \infty$, then renaming (gives $(-z;q)_\infty = \sum q^{\binom{n}{2}} z^n/(q;q)_n$).

9.2 Proof of the finite q-binomial theorem

This proof is purely algebraic (no analysis) and is the version best suited for Lean. It uses only induction on n and the q-Pascal identity.

Lemma 29 (q-Pascal)**.** *$qSeries.qBinom_succ_succdef : qBinomForalln$, $k \in \mathbb{N}$,*

$$\binom{n+1}{k+1}_q = \binom{n}{k+1}_q + q^{n-k}\binom{n}{k}_q.$$

(Equivalently, with $k \geq 1$: $\binom{n+1}{k}_q = \binom{n}{k}_q + q^{n-k+1}\binom{n}{k-1}_q$. A symmetric form $\binom{n+1}{k+1}_q = q^{k+1}\binom{n}{k+1}_q + \binom{n}{k}_q$ also holds; we use the first.)

Proof. This is the recursive defining equation of $\binom{n}{k}_q$ in Definition **??**, hence `rfl` in Lean. Equivalently, by direct computation from the quotient form $\binom{n}{k}_q = (q;q)_n/\big((q;q)_k(q;q)_{n-k}\big)$ (valid when $k \leq n$ and the denominators are nonzero): $\binom{n}{k}_q + q^{n-k+1}\binom{n}{k-1}_q = \binom{n}{k-1}_q \cdot \frac{(1-q^{n-k+1})+q^{n-k+1}(1-q^k)}{1-q^k} = \binom{n}{k-1}_q \cdot \frac{1-q^{n+1}}{1-q^k} = \binom{n+1}{k}_q$, recovering the off-by-one form. □

*Proof of Theorem **??**.* thm:qbin-finitelem:qpascal, lem:qBinom-eq-zero-of-lt, def:qBinom Induction on n, following the Lean proof `qSeries.qBinom_finite_thm` in `Series/qSeries.lean`. The case $n = 0$ gives $1 = 1$.

For the inductive step, set $P_n(z) := \prod_{k=0}^{n-1}(1+zq^k)$ and $Q_n(z) := \sum_{k=0}^{n} q^{\binom{k}{2}}\binom{n}{k}_q z^k$, so the theorem reads $P_n = Q_n$. By `Finset.prod_range_succ` and the inductive hypothesis $P_n = Q_n$,

$$P_{n+1}(z) \;=\; P_n(z)\,(1+zq^n) \;=\; Q_n(z)\,(1+zq^n),$$

so it suffices to show $Q_{n+1}(z) = Q_n(z)\,(1+zq^n)$. We decompose $Q_{n+1}(z) = \sum_{k=0}^{n+1} q^{\binom{k}{2}}\binom{n+1}{k}_q z^k$ in four explicit steps (mirroring `stepB`–`stepD` of the Lean proof).

Step A (peel off $k = 0$ via `Finset.sum_range_succ'`). Using $\binom{n+1}{0}_q = 1$,

$$Q_{n+1}(z) \;=\; 1 \;+\; \sum_{k=0}^{n} q^{\binom{k+1}{2}}\binom{n+1}{k+1}_q z^{k+1}.$$

Step B (q-Pascal split). Apply Lemma **??** inside the sum and split via `Finset.sum_add_distrib`:

$$\sum_{k=0}^{n} q^{\binom{k+1}{2}}\binom{n+1}{k+1}_q z^{k+1} \;=\; \underbrace{\sum_{k=0}^{n} q^{\binom{k+1}{2}}\binom{n}{k+1}_q z^{k+1}}_{S_1} \;+\; \underbrace{\sum_{k=0}^{n} q^{\binom{k+1}{2}+n-k}\binom{n}{k}_q z^{k+1}}_{S_2}.$$

Step C ($S_2 = zq^n\,Q_n(z)$). The identity $\binom{k+1}{2} = \binom{k}{2}+k$ (helper `choose_two_succ`) and $k + (n-k) = n$ (valid because $k \leq n$, by `Nat.sub_add_cancel`) give

$q^{\binom{k+1}{2}+n-k}\, z^{k+1} = z\, q^n \cdot q^{\binom{k}{2}}\, z^k$, hence

$$S_2 \;=\; z\, q^n \sum_{k=0}^{n} q^{\binom{k}{2}} \binom{n}{k}_q z^k \;=\; z\, q^n\, Q_n(z).$$

Step D ($1{+}S_1 = Q_n(z)$)*.* Reindex $j = k{+}1$ to identify $S_1 = \sum_{j=1}^{n+1} q^{\binom{j}{2}} \binom{n}{j}_q z^j$, then adjoin the $j = 0$ term 1 (using `Finset.sum_range_succ'` in reverse):

$$1 + S_1 \;=\; \sum_{j=0}^{n+1} q^{\binom{j}{2}} \binom{n}{j}_q z^j.$$

The $j = n{+}1$ term vanishes by Lemma **??** ($\binom{n}{n+1}_q = 0$), reducing this to $Q_n(z)$.

Combining the four steps, $Q_{n+1}(z) = (1 + S_1) + S_2 = Q_n(z) + zq^n Q_n(z) = Q_n(z)\,(1{+}zq^n)$. The final algebraic identity is closed in Lean by `linear_combination`.

□

9.3 Passing to the infinite case

Proof sketch of Theorem **??***.* The finite form of Theorem **??** is the analogous identity

$$\sum_{k=0}^{n} \frac{(a;q)_k}{(q;q)_k} z^k \cdot \frac{(q^{n-k+1};q)_k}{(q^{n-k+1};q)_k} \;=\; \frac{(az;q)_n}{(z;q)_n} \cdot (\text{correction})$$

— one needs to be careful here; the cleanest finite version is due to Cauchy and reads, for $|z| < 1$,

$$\sum_{k=0}^{n} \binom{n}{k}_q \frac{(a;q)_k}{(q;q)_k} z^k \;\xrightarrow{n\to\infty}\; \sum_{k=0}^{\infty} \frac{(a;q)_k}{(q;q)_k} z^k.$$

The classical route is:

1. Prove the finite identity $(az;q)_n = \sum_{k=0}^{n} \binom{n}{k}_q (-a)^k q^{\binom{k}{2}} z^k \cdot (\text{telescope})$ — a polynomial identity provable by the same induction as Theorem **??**.
2. Pass to the limit $n \to \infty$ using $(q;q)_k/(q;q)_n \to 1/(q^{k+1};q)_\infty$ and dominated convergence (each $|q^{\binom{k}{2}} z^k| \le |z|^k$ is summable when $|z| < 1$).

□

Avoiding the infinite version entirely. For Jacobi's triple product via Zhu's semi-finite proof, you do not actually need Theorem **??**. You only need the *first Euler identity*

$$(-z;q)_\infty = \sum_{n=0}^{\infty} \frac{q^{\binom{n}{2}}}{(q;q)_n} z^n,$$

which is the $n \to \infty$ limit of Theorem **??** after rewriting $\prod_{k=0}^{n-1}(1 + zq^k)$ as $(-z;q)_n$ and dividing by suitable factors. This limit is gentler than the full Cauchy identity.

9.4 Formalization plan in Lean

Definitions.

- `qPochhammer (a q : R) : ℕ → R`, defined recursively:
  `qPochhammer a q 0 = 1`,
  `qPochhammer a q (n+1) = qPochhammer a q n * (1 - a * q^n)`.
  Use `Polynomial.eval` for the abstract version in $R[X]$, or work directly in $\mathbb{C}$.

- `qBinom (n k : ℕ) (q : R) : R` as `qPochhammer q q n / (qPochhammer q q k * qPochhammer q q (n - k))` when $k \leq n$, else 0. To stay in $\mathbb{Z}[q]$, define instead as a polynomial via the q-Pascal recurrence and prove the closed form is well-defined.

Recommended approach: define q-binomial coefficients via the recurrence, not the quotient. The quotient definition introduces division-by-zero conditions (the denominator $(q;q)_k$ vanishes at roots of unity). The recurrence-based definition avoids this entirely:

> `qBinom 0 0 q = 1`; `qBinom 0 (k+1) q = 0`; `qBinom (n+1) 0 q = 1`;
> `qBinom (n+1) (k+1) q = qBinom n (k+1) q + q^(n-k) * qBinom n k q`.

Then `qBinom n k` is a polynomial in q with integer coefficients by construction. The closed-form equality with $(q;q)_n/((q;q)_k(q;q)_{n-k})$ becomes a *theorem* provable for q outside the relevant roots of unity (or as an identity of rational functions).

Step-by-step Lean roadmap.

1. Define `qPochhammer` (Mathlib's `Finset.prod_range` suffices: `qPochhammer a q n := ∏ k ∈ range n, (1 - a * q^k)`). ~ 30 **lines.**

2. Prove basic identities: `qPochhammer a q (n+1) = qPochhammer a q n * (1 - a * q^n)` (this is `Finset.prod_range_succ`); the recursion the other way; multiplicativity. ~ 50 **lines.**

3. Define `qBinom` by recursion as above; prove symmetry $\binom{n}{k}_q = \binom{n}{n-k}_q$ by induction. ~ 100 **lines.**

4. Prove the closed-form formula $\binom{n}{k}_q \cdot (q;q)_k \cdot (q;q)_{n-k} = (q;q)_n$ as an identity of polynomials. This is the bridge between the recursive definition and the quotient definition. ~ 100 **lines.**

5. Prove the finite q-binomial theorem (Theorem **??**) by induction, following the proof above. ~ 80 **lines.**

6. Define `qPochhammerInf` as a `tprod`; prove `Multipliable` for $|q| < 1$ using $\sum \|q^k\| < \infty$. ~ 100 **lines.**

7. Prove the first Euler identity $(-z; q)_\infty = \sum q^{\binom{n}{2}} z^n/(q; q)_n$ for $|q| < 1$ and z in any bounded set, by taking $n \to \infty$ in Theorem **??**. The convergence step uses dominated convergence with the bound $|q^{\binom{n}{2}}| \le |q|^{n(n-1)/2}$. $\sim$ 200 **lines.**

8. (Optional) Prove the full Cauchy identity (Theorem **??**). ~ 150 **extra lines.**

Total for steps 1–7 (sufficient for Zhu's proof of Jacobi): ~ 650 lines. Adding Zhu's argument (~ 100 lines) plus the pentagonal corollary (~ 60 lines) brings the total Jacobi-via-q-binomial route to ~ 800 lines plus whatever scaffolding the bilateral series on the RHS requires.

9.5 Existing Mathlib infrastructure

Algebraic foundations.

- `Mathlib.Algebra.BigOperators.Basic`, `Mathlib.Algebra.BigOperators.Ring` – the `∏ k ∈ s, f k` and `∑ k ∈ s, f k` machinery, including `Finset.prod_range_succ`, `Finset.prod_range_succ'`, `Finset.prod_Ico_consecutive`, `Finset.prod_bij`.

- `Mathlib.Algebra.Polynomial.Basic` – if you want to formalize q-binomials as elements of $\mathbb{Z}[q]$ rather than as functions of q, this gives `Polynomial`, `Polynomial.coeff`, `Polynomial.eval`.

- `Mathlib.Algebra.Polynomial.Induction` – `Polynomial.induction_on'` for proving polynomial identities by reducing to monomials.

- `Mathlib.Combinatorics.Choose.Basic` – ordinary binomial coefficients `Nat.choose`, useful as a guide and for the $q \to 1$ limit (not needed for the proof itself).

Analytic foundations (for the infinite case).

- `Mathlib.Topology.Algebra.InfiniteSum.Defs` – `Summable`, `Multipliable`, `tsum` ($\sum'$), `tprod` ($\prod'$).

- `Mathlib.Analysis.Normed.Field.InfiniteSum` – `Summable.mul_norm`, Cauchy products, dominated convergence for sums.

- `Mathlib.Analysis.SpecificLimits.Normed` – `summable_geometric_of_norm_lt_one`, the workhorse summability lemma.

- `Mathlib.Analysis.Normed.Group.InfiniteSum` – `tsum_geometric_of_norm_lt_one` and friends.

What is missing (as of early 2026).

- No `qPochhammer` or `qPochhammerInf`.
- No `qBinom` or Gaussian binomial coefficient. (Mathlib's existing `pochhammer` is the *rising factorial* $x(x+1)\cdots(x+n-1)$, a different and unrelated object despite the name.)
- No q-Pascal recurrence, no finite or infinite q-binomial theorem, no Euler identities.

This means the entire q-series scaffold needs to be developed from scratch. The good news is that there are no foundational gaps: every step rests on Mathlib infrastructure that already exists. The work is purely additive.

9.6 Why this is worth it

Once the ~ 650 lines of q-series infrastructure are in place, the following all become short corollaries:

- Jacobi's triple product (Zhu's proof: ~ 100 lines).
- Euler's pentagonal number theorem (specialization: ~ 60 lines).
- Gauss's identities for $\sum q^{n(n+1)/2}$ and $\sum(-1)^n q^{n^2}$ (specializations: ~ 50 lines each).
- The generating function for partitions $\sum p(n)q^n = 1/(q;q)_\infty$ (immediate).
- Generating functions for partitions into distinct parts, odd parts, etc. (Euler's theorem: ~ 100 lines).

By contrast, formalizing any one of these via a bespoke proof would cost ~ 1000–2000 lines and produce no reusable infrastructure. The q-series route amortizes the cost.

10 The q-Pochhammer symbol and Gaussian binomial coefficient (formalized)

This section records the foundational definitions and lemmas that have been formalized in Lean for the q-series approach to Jacobi's triple product. The corresponding source file is `Series/qSeries.lean` in the `Series` Lean library, under the namespace `qSeries`. Throughout this section R is a commutative ring and $a, q \in R$.

10.1 The q-Pochhammer symbol

Definition 30 (Finite q-Pochhammer symbol)**.** qSeries.qPochhammerFor $n \in \mathbb{N}$, the finite q-Pochhammer symbol is

$$(a;q)_n \; := \; \prod_{k=0}^{n-1} \bigl(1 - a\,q^k\bigr).$$

In Lean, this is implemented as `∏ k ∈ Finset.range n, (1 - a * q^k)`, specialising to $(a;q)_0 = 1$ (the empty product).

Lemma 31 (Base case)**.** *qSeries.qPochhammer*$_z$*erodef : qPochhammer (a; q)*$_0 = 1$.

Proof. The empty product is 1 (`Finset.prod_range_zero`). □

Lemma 32 (Recurrence)**.** *qSeries.qPochhammer*$_s$*uccdef : qPochhammerForalln* $\in \mathbb{N}$*,*

$$(a;q)_{n+1} \; = \; (a;q)_n \cdot \bigl(1 - a\,q^n\bigr).$$

Proof. Immediate from `Finset.prod_range_succ`. □

10.2 The Gaussian binomial coefficient

Definition 33 (Gaussian binomial coefficient)**.** qSeries.qBinomThe Gaussian binomial coefficient $\binom{n}{k}_q \in R$ is defined by recursion on n and k:

$$\begin{aligned} \binom{0}{0}_q &= 1, \quad \binom{0}{k+1}_q = 0, \\ \binom{n+1}{0}_q &= 1, \quad \binom{n+1}{k+1}_q = \binom{n}{k+1}_q + q^{n-k}\binom{n}{k}_q. \end{aligned}$$

This is the q-Pascal recurrence in the form stated in Lemma **??** (with the index shift $k \mapsto k+1$). By construction $\binom{n}{k}_q$ is a polynomial in q with integer coefficients, defined for every pair $(n,k) \in \mathbb{N}^2$ (no division, no roots-of-unity exclusion).

Lemma 34 (Vanishing above the diagonal)**.** $qSeries.qBinom_e q_z ero_o f_l tdef : qBinomIfn < k, then \binom{n}{k}_q = 0.$

Proof. Induction on n and k following the four-case recursive definition. The non-trivial case is $(n+1, k+1)$ with $n < k$: the recurrence gives $\binom{n+1}{k+1}_q = \binom{n}{k+1}_q + q^{n-k}\binom{n}{k}_q$, and both inductive hypotheses (at $(n, k+1)$ and (n,k), available because $n < k+1$ and $n < k$ respectively) yield 0. □

Lemma 35 (Diagonal)**.** $qSeries.qBinom_s elfdef : qBinom, lem : qBinom - eq - zero - of - lt\binom{n}{n}_q = 1$ *for every* $n \in \mathbb{N}$*.*

Proof. Induction on n. The case $n = 0$ is by definition. For the step,

$$\binom{n+1}{n+1}_q = \binom{n}{n+1}_q + q^{n-n} \binom{n}{n}_q = 0 + 1 \cdot 1 = 1,$$

using Lemma **??** for the first term and the inductive hypothesis for the second. □

Lemma 36 (Closed form for $\binom{n}{k}_q$). *$qSeries.qBinom_mul_qPochhammer_mul_qPochhammerdef : qBinom, def : q$ $\leq n$,*

$$\binom{n}{k}_q \cdot (q;q)_k \cdot (q;q)_{n-k} \ = \ (q;q)_n.$$

Proof. Induction on n.

Base case $n = 0$. Forces $k = 0$ and the identity is $1 \cdot 1 \cdot 1 = 1$.

Boundary $k = 0$. Reduces to $1 \cdot 1 \cdot (q;q)_n = (q;q)_n$.

Boundary $k = n + 1$. By Lemma **??** we have $\binom{n+1}{n+1}_q = 1$, and $(q;q)_0 = 1$, so both sides equal $(q;q)_{n+1}$.

Inductive step $k + 1 \leq n$. Apply the q-Pascal recurrence

$$\binom{n+1}{k+1}_q = \binom{n}{k+1}_q + q^{n-k} \binom{n}{k}_q$$

and the factorisations

$$(q;q)_{n-k} \ = \ (q;q)_{n-(k+1)} \cdot \bigl(1 - q\, q^{n-(k+1)}\bigr), \qquad (q;q)_{k+1} \ = \ (q;q)_k \cdot \bigl(1 - q\, q^k\bigr),$$

together with the power identities $q \cdot q^{n-(k+1)} = q^{n-k}$ and $q^{n-k} \cdot (q \cdot q^k) = q \cdot q^n$, which use $k \leq n$ via `Nat.sub_add_cancel`. The two inductive hypotheses at $(n, k+1)$ and (n, k) then combine into the required identity, dispatched in Lean by `linear_combination`. □

10.3 The finite q-binomial theorem (step 5)

Theorem **??** is formalised as `qSeries.qBinom_finite_thm` in `Series/qSeries.lean`. The Lean proof follows exactly the four-step strategy spelled out in the proof attached to Theorem **??** above: peel off $k = 0$, apply the q-Pascal recurrence (Lemma **??**) inside the sum, simplify the second piece to $zq^n\, Q_n$, and recombine the first piece via $\binom{n}{n+1}_q = 0$ (Lemma **??**).

10.4 The infinite q-Pochhammer symbol and Cauchy identity (steps 6, 8)

Working in $\mathbb{C}$ for the analytic part of the development.

Definition 37 (Infinite q-Pochhammer symbol)**.** qSeries.qPochhammerInf

$$(a;q)_\infty \;:=\; \prod_{k=0}^{\infty} \bigl(1 - a\,q^k\bigr) \;\in\; \mathbb{C},$$

defined unconditionally via `tprod`. The value is mathematically meaningful (and the product is convergent) under the hypothesis $\|q\| < 1$ provided by Lemma **??**.

Lemma 38 (Multipliability)**.** *$qSeries.multipliable_{o}ne_{s}ub_{s}mul_{q}powdef : qPochhammerInfFor a,$ $q \in \mathbb{C}$ with $\|q\| < 1$, the family $k \mapsto 1 - a\,q^k$ is multipliable; in particular the partial products converge to $(a;q)_\infty$.*

Proof. Reduce multipliability to summability of $\sum_k \|aq^k\|$ via Mathlib's `multipliable_one_add_of_summable` (which works in any complete normed ring), then bound $\|aq^k\| = \|a\|\,\|q\|^k$ and conclude by the geometric series $\sum_k \|q\|^k < \infty$ (`summable_geometric_of_lt_one`). □

Theorem 39 (Infinite q-binomial / Cauchy identity, restated for the formal development)**.** *$qSeries.qBinom_{i}nfinite_{t}hmdef : qPochhammer, def : qPochhammerInf, lem : multipliable - o$ $z,\ q \in \mathbb{C}$ with $\|q\| < 1$ and $\|z\| < 1$,*

$$\mathrm{HasSum}\Bigl(n \mapsto \tfrac{(a;q)_n}{(q;q)_n}\,z^n,\ \tfrac{(az;q)_\infty}{(z;q)_\infty}\Bigr).$$

*(Equivalent to Theorem **??**.) The Lean statement is `qSeries.qBinom_infinite_thm` in `Series/qSeries.lean`; the proof is currently a stub (`sorry`). The standard route is to take $n \to \infty$ in the finite identity (a Cauchy-type analog of Theorem **??**), applying dominated convergence on the summable side and multipliability (Lemma **??**) on the product side.*

Remark. Together with Lemma **??**, Lemma **??**, and Lemma **??**, Theorem **??** realises steps 1, 2, 3, 4, and 5 of the formalisation roadmap in Section **??** ("Formalization plan in Lean"). Step 6 is realised by Definition **??** and Lemma **??**. Step 8 (the full Cauchy identity, Theorem **??**) has its statement formalised (`qSeries.qBinom_infinite_thm`) but its proof remains as future work. Step 3 is partially completed: the recursive definition is in place (Definition **??**) and the closed-form bridge to the quotient description is established (Lemma **??**); the symmetry identity $\binom{n}{k}_q = \binom{n}{n-k}_q$ remains as future work, as does step 7 (the first Euler identity).